

Astronomical spectroscopy I.



Topics

1. The history of stellar spectroscopy
2. The properties of spectroscopes, resolution, spectral range, etc.
3. Convolution and Fourier-transformation
4. Spectroscope-types: prism, Cassegrain-, echelle-, Coudé spectroscopes
5. Multi-object spectrographs and Fourier-transform spectrometer
6. Reducing Cassegrain spectra in IRAF
7. Wavelength calibration, continuum normalization
8. Measuring radial velocities, cross-correlation
9. Reducing Echelle spectra
10. Flux calibration and measuring equivalent widths
11. Measuring abundances from stellar spectra

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Requirements

- **It is a must to attend the course!** The attendance will be booked in Neptun at the beginning of each lecture (in other words we catalogue who is present).
- The course contains 12 lectures. **Students are allowed to be absent from 3 lectures in total.** Students who are absent from 4 or more lectures will not get a grade at the end of the course! They will need to repeat the course in a different semester.
- Students who missed a course need to do/present the related tasks at the beginning of the next lecture. Tasks made at home may be acceptable, but only when someone misses a lecture.
- **Students can be late from the lecture by a maximum of 10 minutes.** Arrival 10 minutes after the lecture starts will be automatically counted as being absent.
- Student will receive their grade (from 1 to 5) in the 12th lecture. There will be no ZH during this course. **Students get their grade for what tasks they successfully finish and present in the entire course;** this includes carrying out IRAF, normplot and Synspec related exercises. Tasks made at home will not be accepted (for exceptions see previous point)!
- All lecture material will be available in pdf form from Canvas.
- If the lecture is delayed due to the lecturer being absent, all students will be counted as if they attended the lecture.

Installation of iraf

Linux in Windows: <https://learn.microsoft.com/en-us/windows/wsl/install>

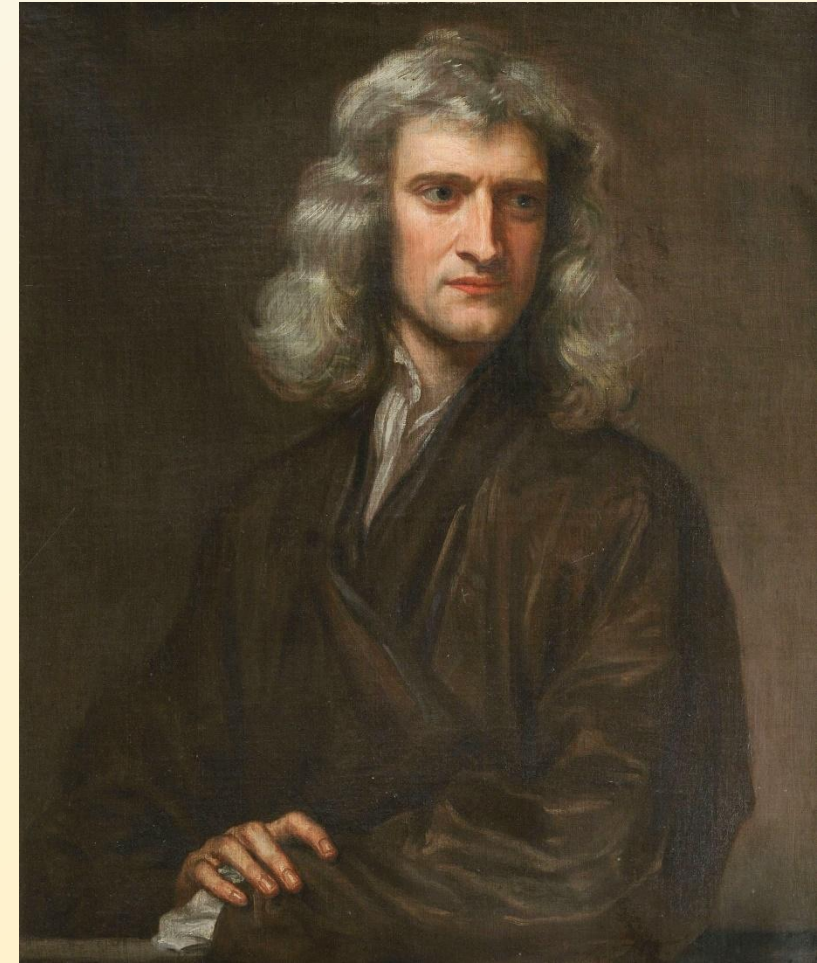
Iraf: <https://iraf-community.github.io/install.html>

Follow these steps:

1. Start Ubuntu from the Windows Start Menu
2. In the Ubuntu terminal window type *sudo apt get update*
3. The sudo command will ask for your password, so type it in and press enter.
4. Then type *sudo apt install iraf xterm*
5. Now you'll see a whole load of text
6. Press enter to accept the default (capitalized) answer of "Y"
7. Watch as it downloads and installs packages
8. When it's done type *xterm* & and press enter
9. In the xterm window type *irafcl* and press enter
10. In linux you may need to run *mkiraf* first and then run *cl* to start iraf.

Isaac Newton (1642-1726)

- 1666: he let a single narrow beam of sunlight through a hole, which passed into the prism.
- He observed no absorption lines, because of low resolution and using a circular hole instead of a slit.
- Different colors correspond to different refraction indexes.



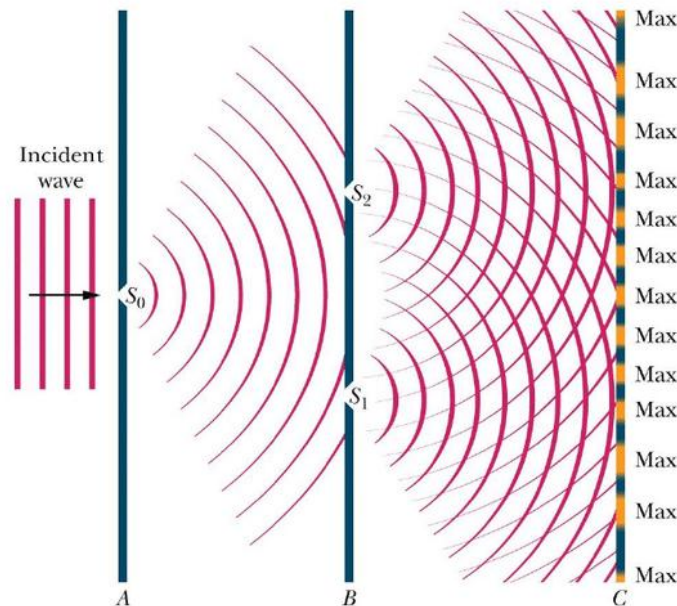
William Herschel (1738 - 1822)

- 1800: different colors generate varying amounts of heat.
- He detected heat from light beyond the red color.
- He concluded that there must be an invisible form of light beyond the visible spectrum.

Thomas Young (1773-1829)

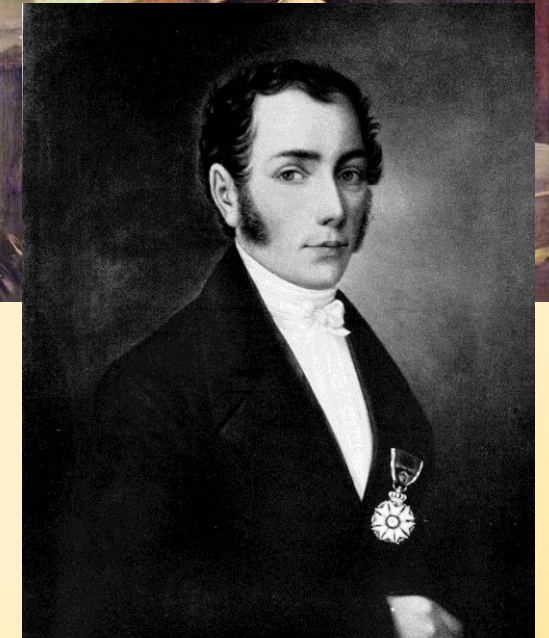
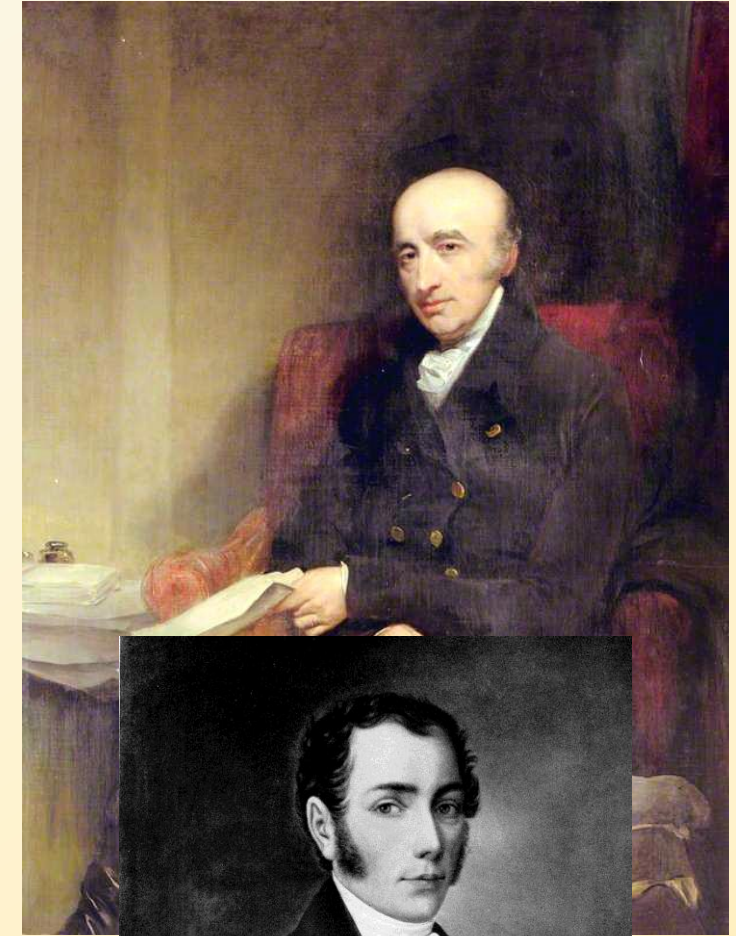
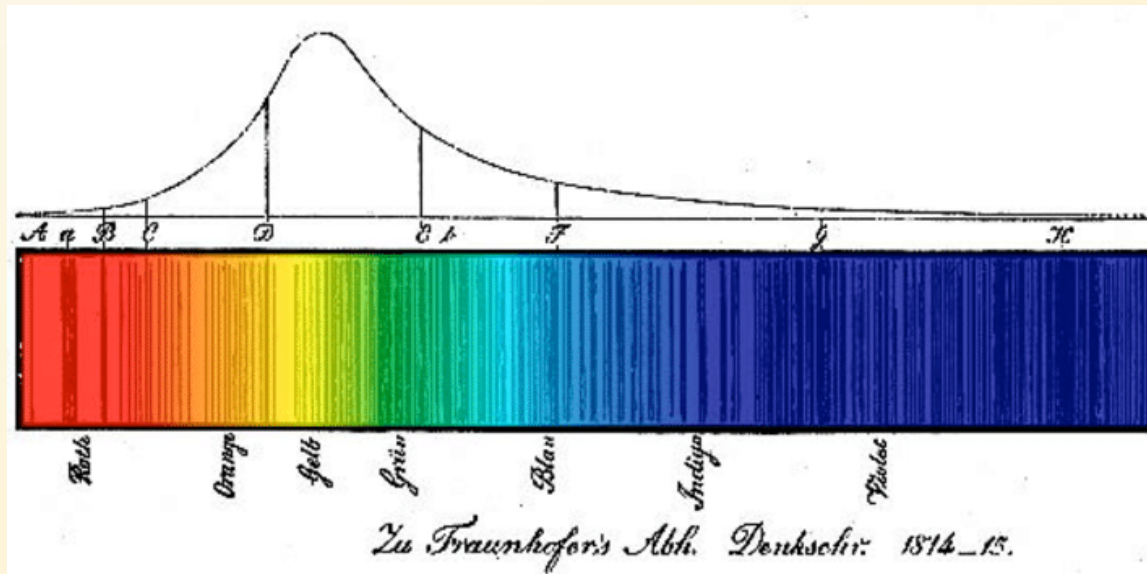
- 1800: diffraction and interference of light.
- He used an optical grating, tiny grooves in material.
- He concluded that light is composed of waves.

6. The figure shows an outline of Young's double slit experiment. Describe your thoughts and reasons for the areas where the dark patterns appear in terms of energy.



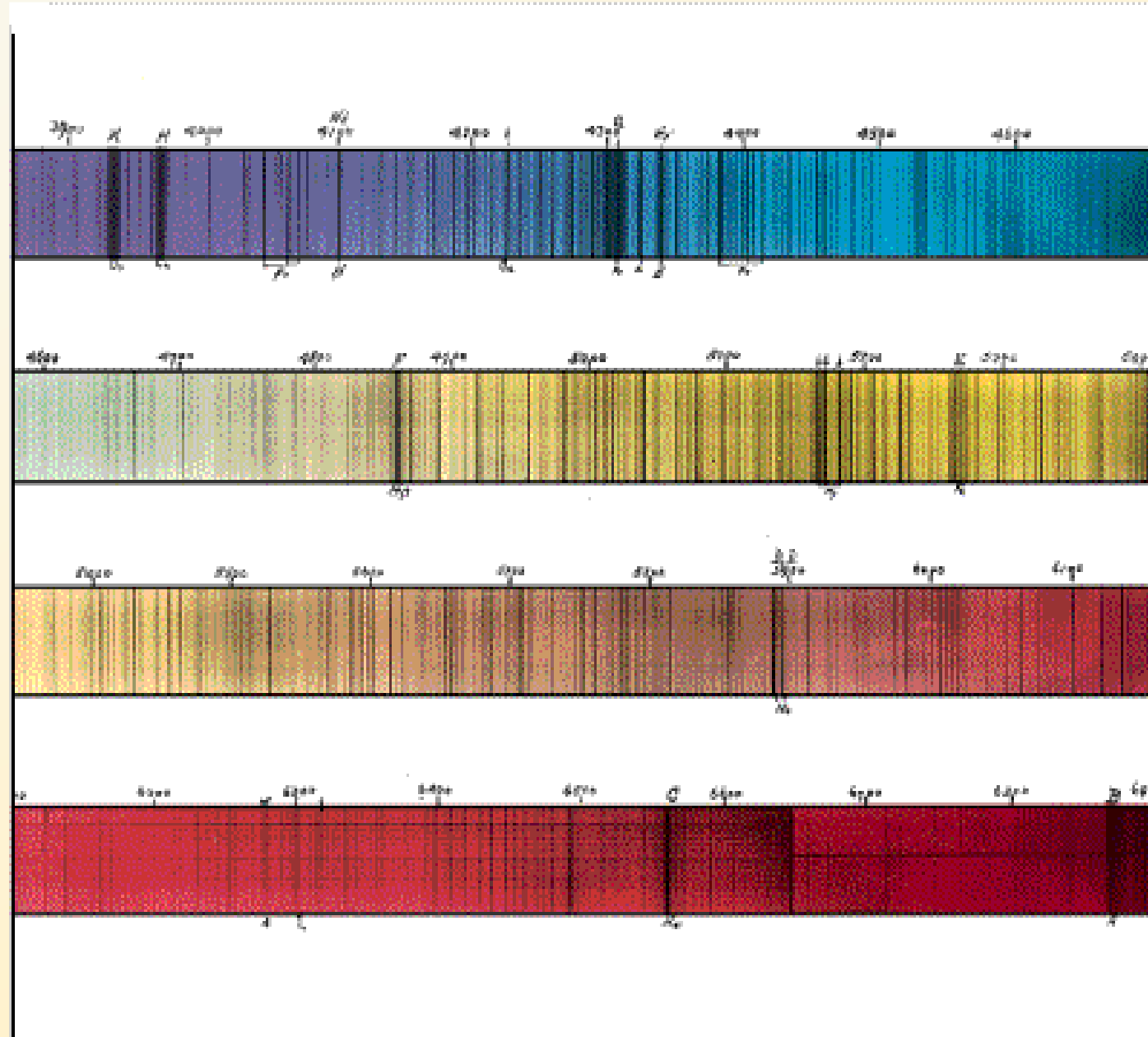
Discovery of dark lines in the Solar spectrum

- **William Wollaston (1766-1828)**, 1802: observations of dark gaps in the solar spectrum, he discovered 7 lines using a narrow slit for higher resolution.
- **Joseph Fraunhofer (1787 - 1826)**
- 1814: he builds a spectroscope.
- The dark lines do not mark the edge of colors.
- He identified 574 lines, marked 10 by A-K letters.

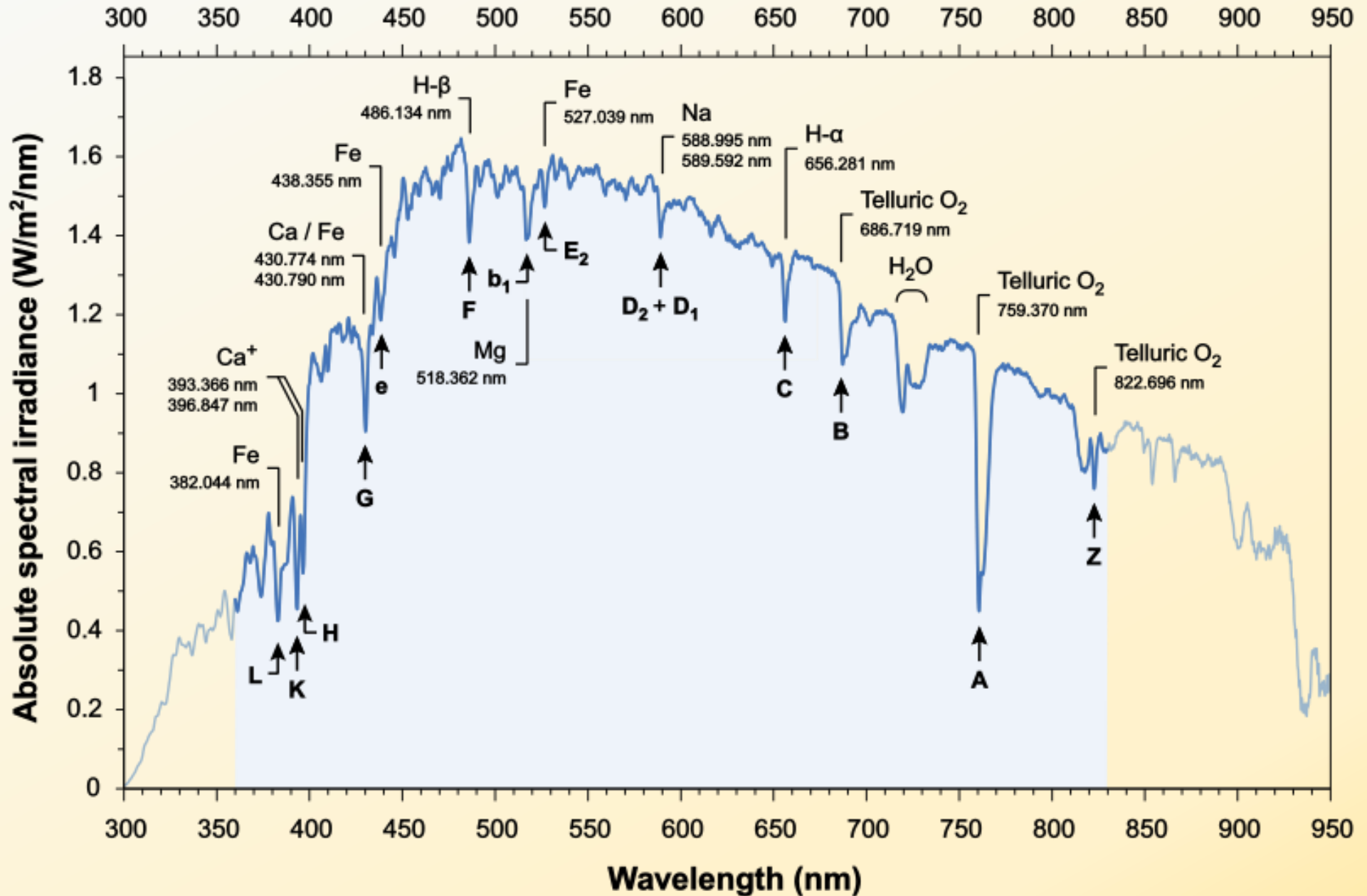


Fraunhofer lines

- He measured the position of 350 lines.
- Venus has similar lines to the Sun.
- Sirius has three wide lines, it is different from the Sun.
- He observed dark lines in bright stars, but at different positions.
- Lines originate from the stars and not from the Earth's atmosphere.

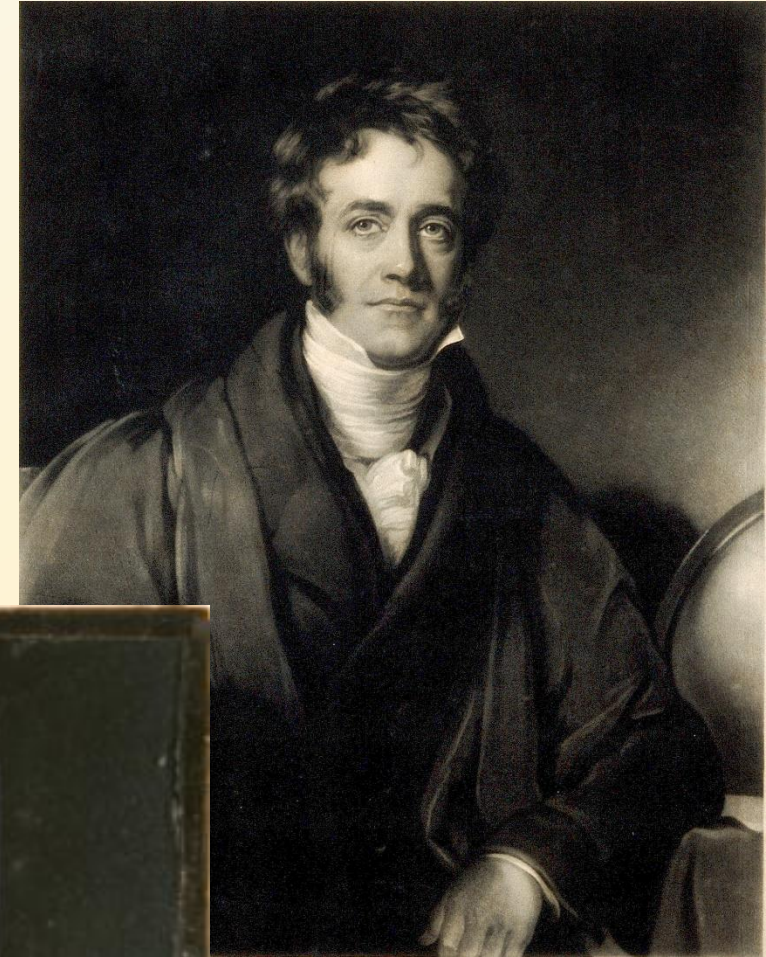


Fraunhofer lines



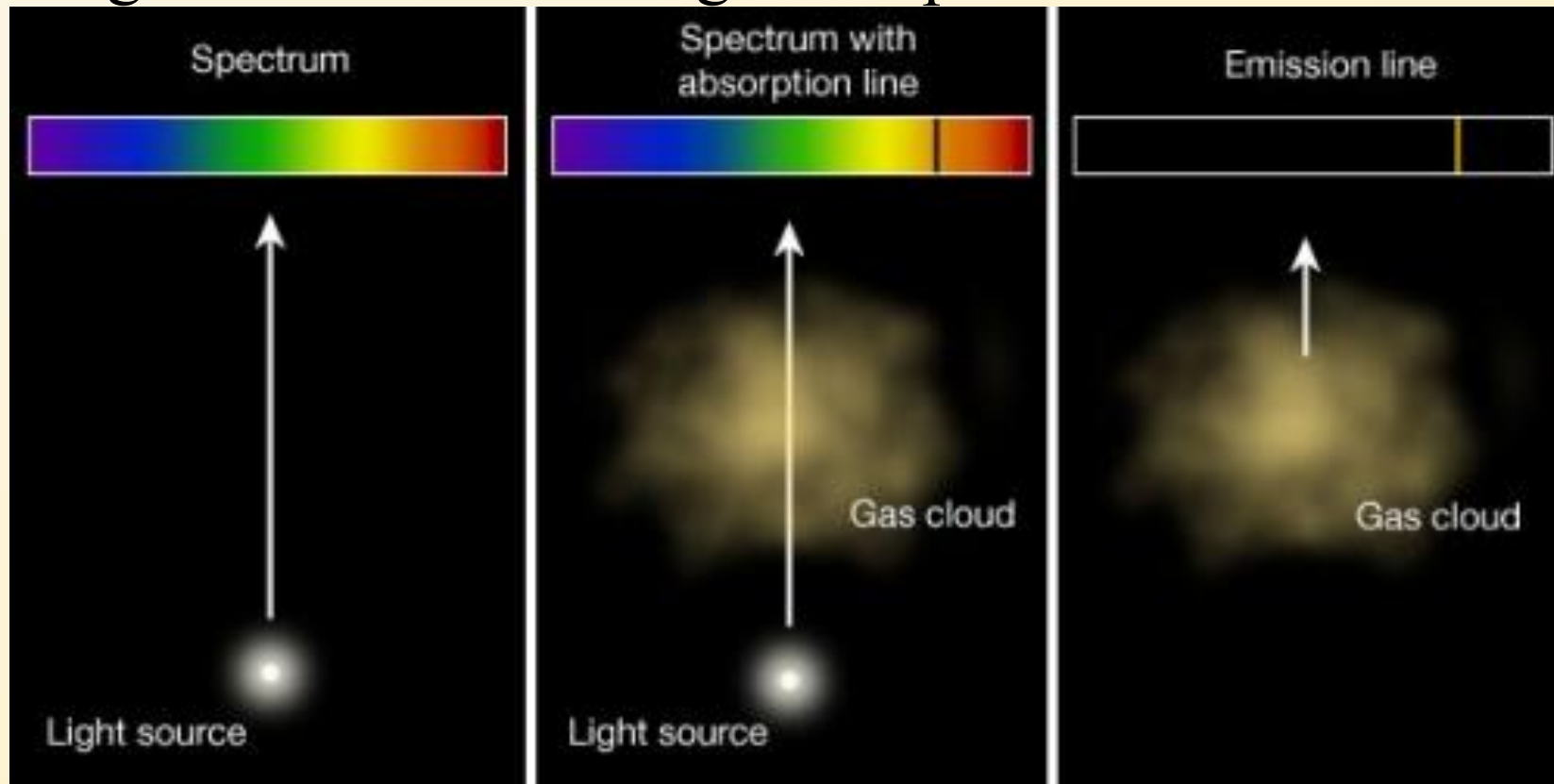
Obtaining the first spectra of objects

- **William Herschel (1792 - 1871)**
- Takes the spectra of various flames.
- Obtains infrared spectra in photographic plates.
- Rays of different parts of the spectrum impart their own color to a photographic paper.
- **Edmond Becquerel (1820-1891)**
- 1848: takes the first photo of the Sun's spectrum.
- Discovered the first UV lines.



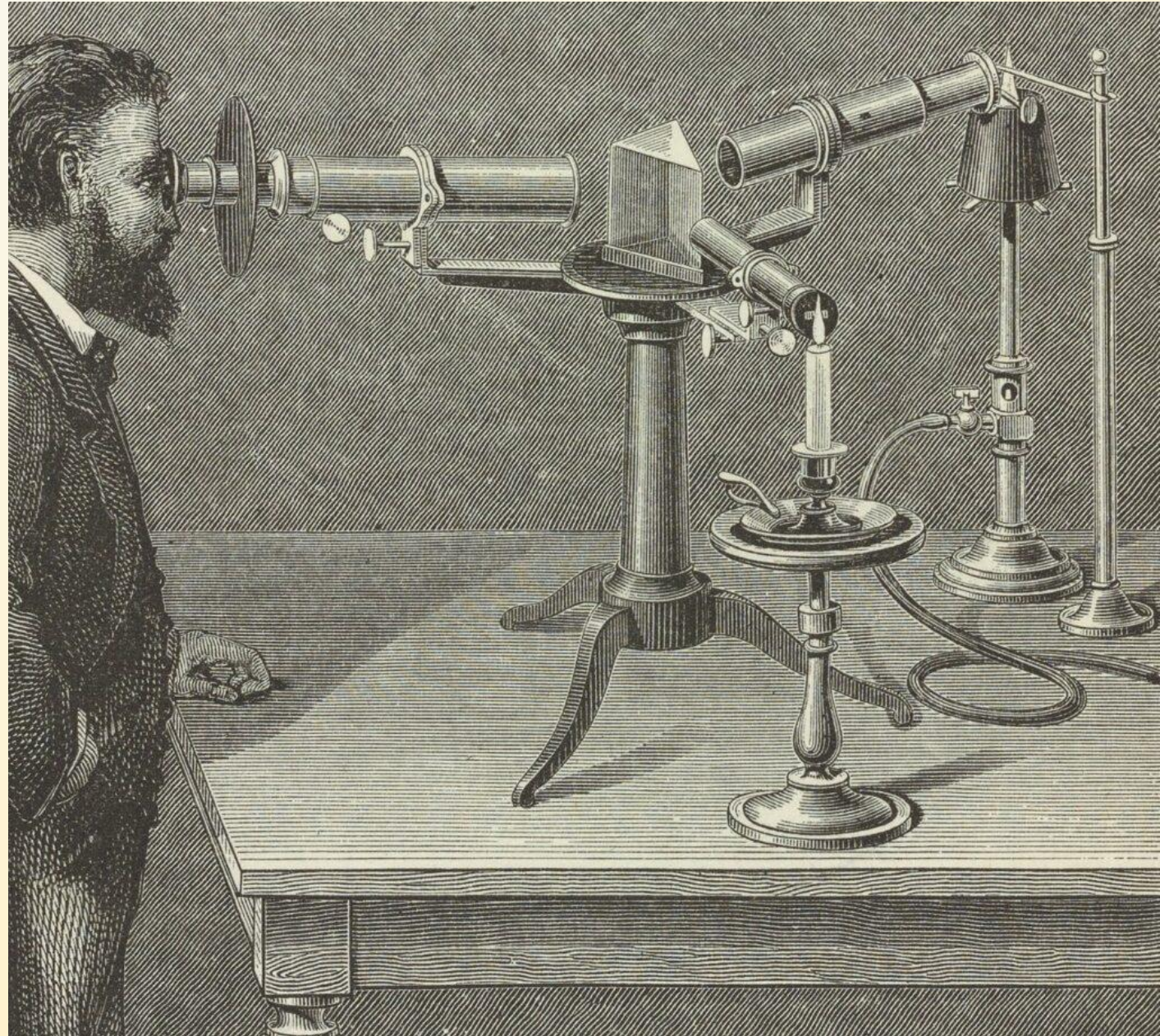
Observing the spectra of objects

- **Gustav Kirchhoff (1824-1887)**
- The Na D emission and absorption lines have the same wavelength.
- A luminous solid, liquid, or dense gas emits light of all wavelengths.
- A low density, hot gas seen against a cooler background produces an emission line spectrum.
- A low density, cool gas in front of a hotter source of a continuous spectrum creates absorption line spectrum.



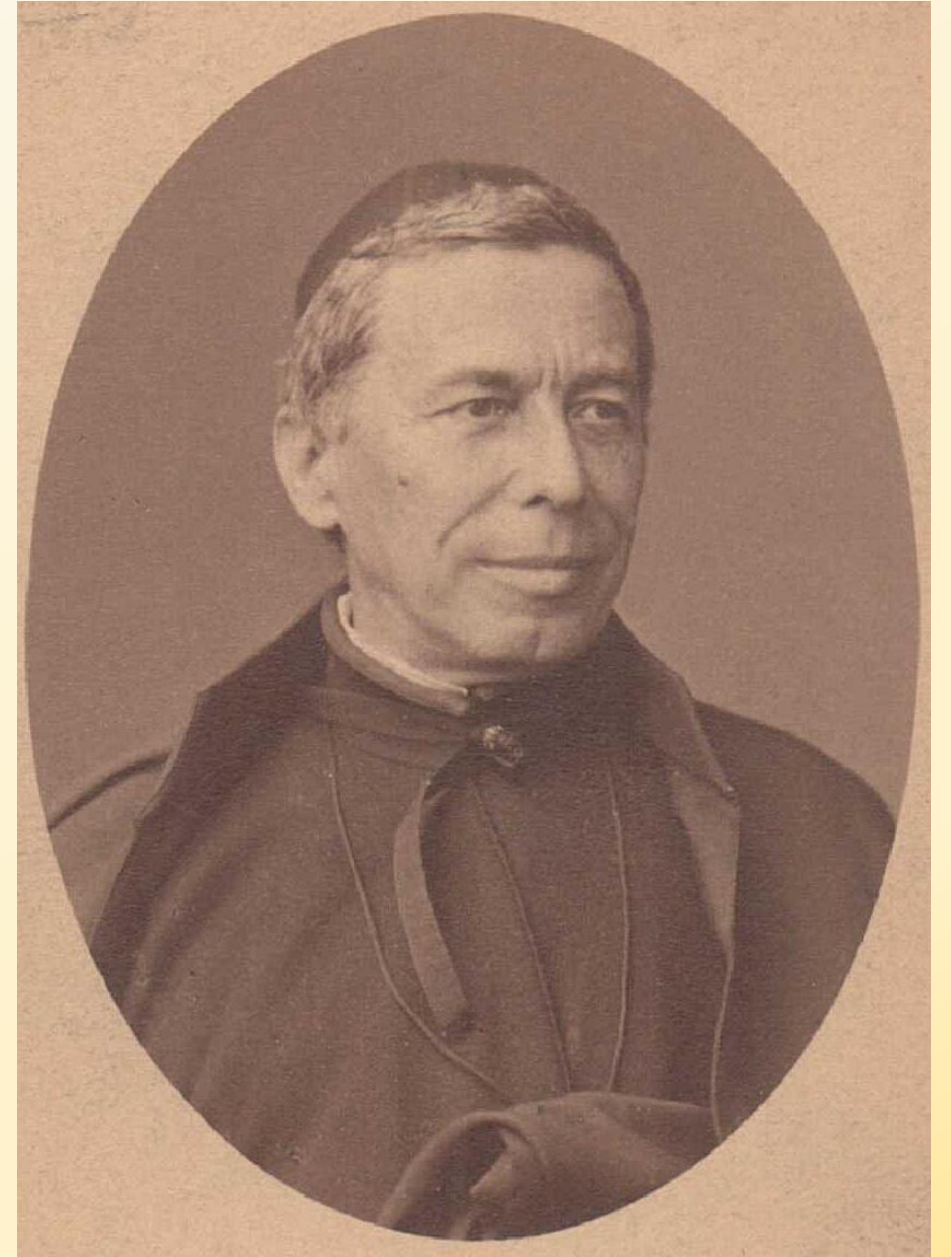
Observing the spectra of objects

- **Robert Bunsen (1811-1899)**
- With Kirchhoff they did chemical analysis by identifying spectral lines.
- Conclusion: in order to identify the lines in the Sun's spectrum, one has to observe the lines of various materials in the laboratory.
- Kirchhoff identified multiple elements in the Sun's spectrum: Fe, Ca, Mg, Na, Ni, Cr.



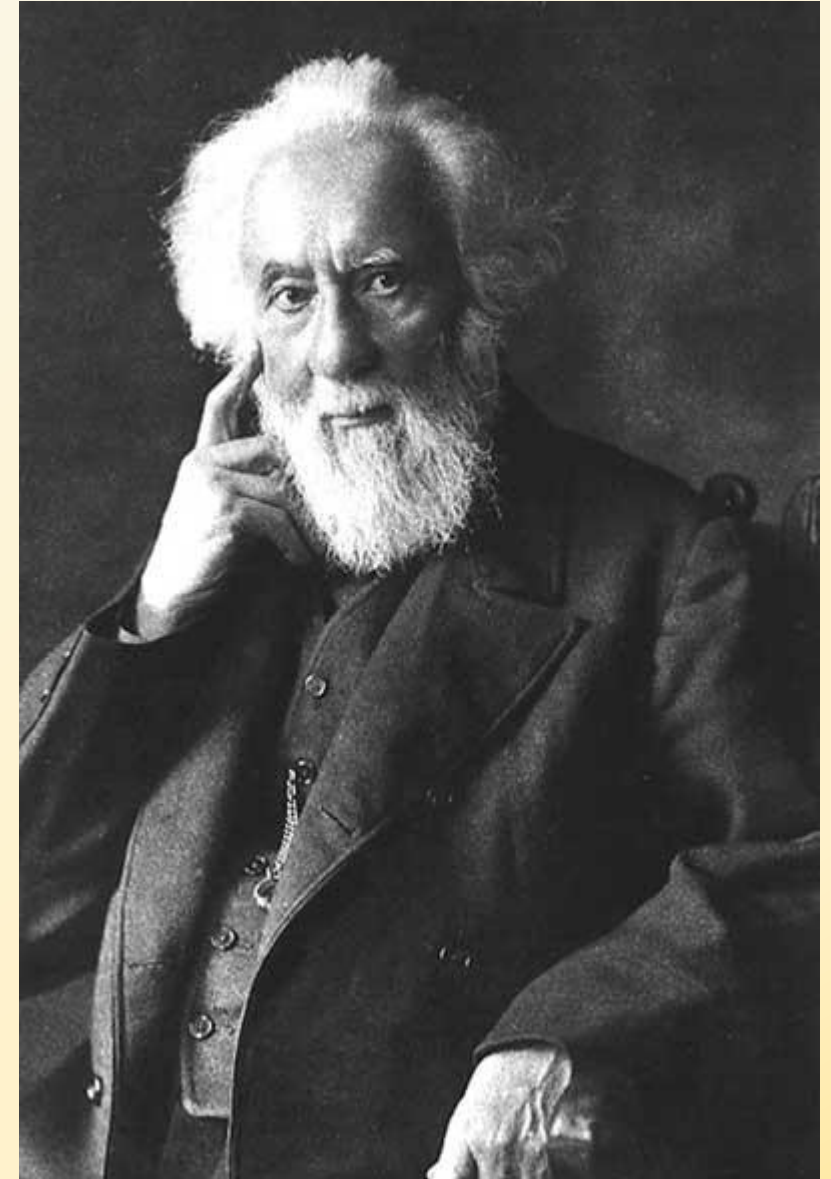
Observing the spectra of objects

- **Angelo Secchi (1818-1878)**
- Priest and an astronomer.
- The Sun is a star, not a planet!
- Starting in 1863 he accumulated 4,000 stellar spectrograms.
- Stars have a limited number of distinct types and subtypes, which could be distinguished by their different spectral patterns → 5 types called Secchi classes.
- Discovered stars rich in carbon.
- He wrote 700 articles, 2 books in 30 years.



Observing the spectra of objects

- **William Huggins (1842-1910)**
- 1862 and onwards: he compared spectra of stars with spectral lamps in the laboratory.
- Identified various chemical elements in stellar spectra.
- 1864: he took spectra of nebulae, which showed emission lines, so he concluded they are not stars.
- Identified CN in the spectrum of the Tempel comet.
- 1866: took the spectrum of novae, they have both emission and absorption lines.
- 1868: measured the Doppler-shift of the spectrum of Sirius.



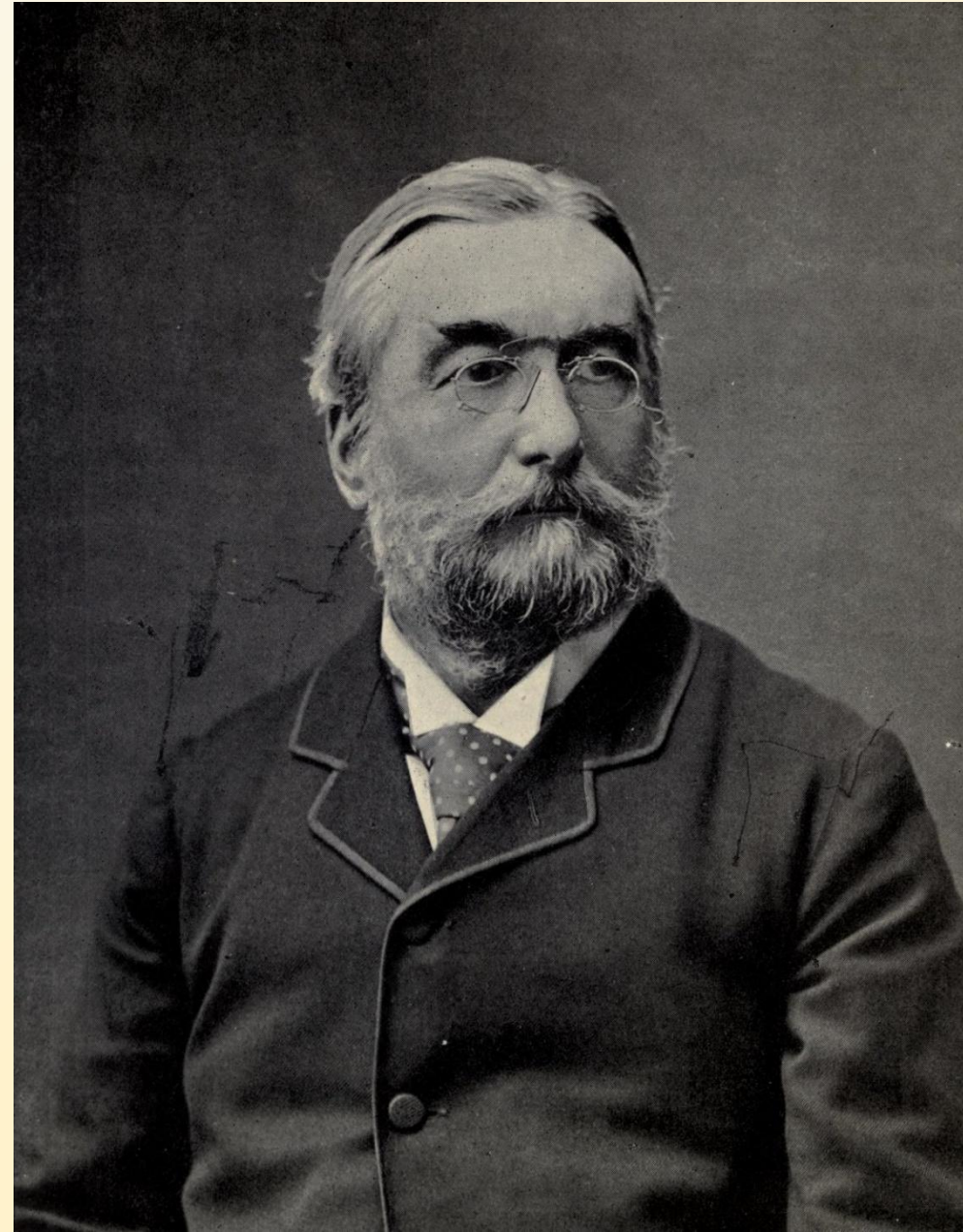
Observing the spectra of objects

- **Hermann Carl Vogel (1841-1907)**
- Spectra of certain stars shifted slightly over time, moving toward the red and then later toward the blue.
- First to establish Sun's rotational period using the Doppler effect.
- Measured the radial velocity of stars with a precision of about 3 km/s.
- Took the radial velocity curve of the first spectroscopic binary: Algol.
- Discovery of the first one line spectroscopic binary: α Vir.
- 3 spectral types: white, yellow and red.



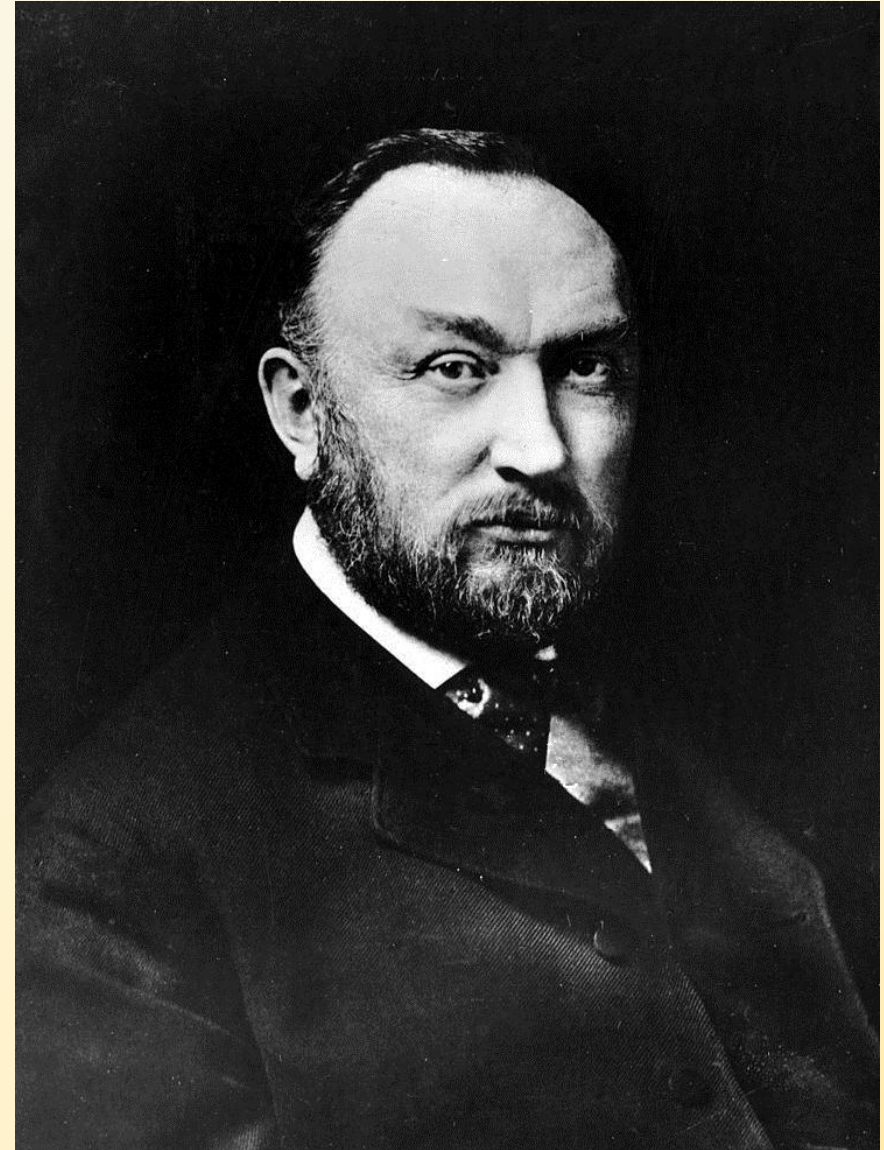
Observing the spectra of objects

- **Norman Lockyer (1836-1920)**
- Founded the journal Nature in 1869.
- 1868: discovered the first element unknown to mankind in the spectrum of the Sun, helium.
- The lines was at 588 nm, slightly bluer than the Na D lines.
- He differentiated dwarf stars from giants.
- His theory was that stars evolve from red to white to red again.



Stellar classification

- **Edward C. Pickering (1846-1919)**
- Director of Harvard College Observatory (HCO) from 1877 to his death in 1919.
- 1882: developed a method to photograph the spectra of multiple stars simultaneously by putting a large prism in front of the photographic plate.
- He and his team captured images of over 220,000 stars.
- Pickering recruited over 80 women to work for him, including Annie Jump Cannon, Antonia Maury, and Williamina Fleming.



Stellar classification

- **Williamina Fleming (1857 – 1911)**
- She cataloged 26,000 stars into 13 different classes based on their spectra.
- **Antonia Maury (1866 - 1952)**
- She identified 22 different classes.
- Discovery of the second spectroscopic binary:
 β Aurigae.
- They used Balmer lines:
A had the strongest,
B the second strongest,
and so on...



Harvard computers (Pickering's Harem)



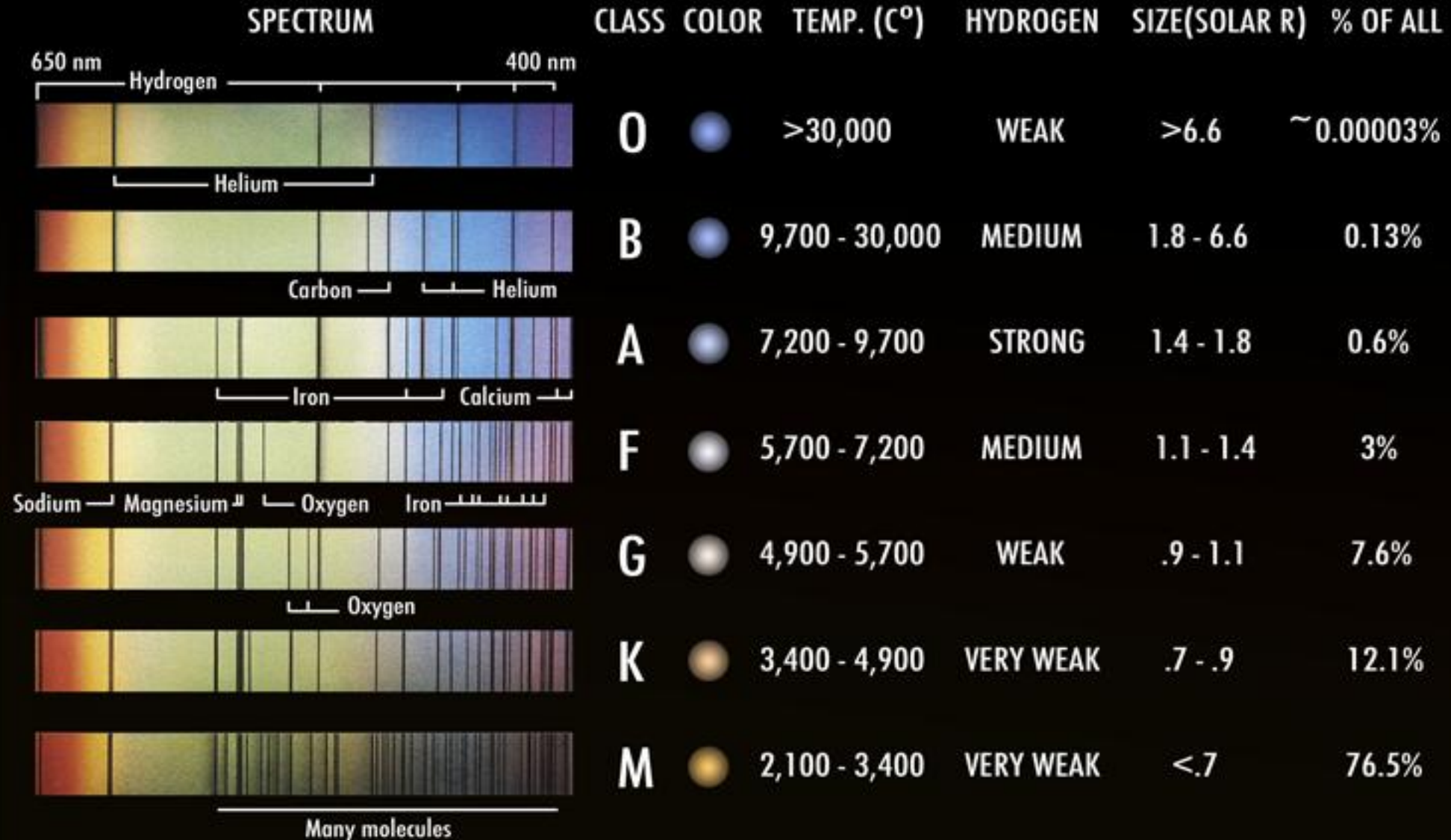
Spectral classification

- **Annie Cannon (1863-1941)**
- She created the Harvard Classification Scheme that organizes spectral types based on temperature and is still in use today.
- Spectral classes: O, B, A, F, G, K, M, based on the strength of the Balmer lines.
- Each divided into 10 subgroups.
- She classified 350,000 stars in her lifetime!
- They created the Henry Deaper Catalogue: classification of 225,300 stars brighter than 9 magnitude.

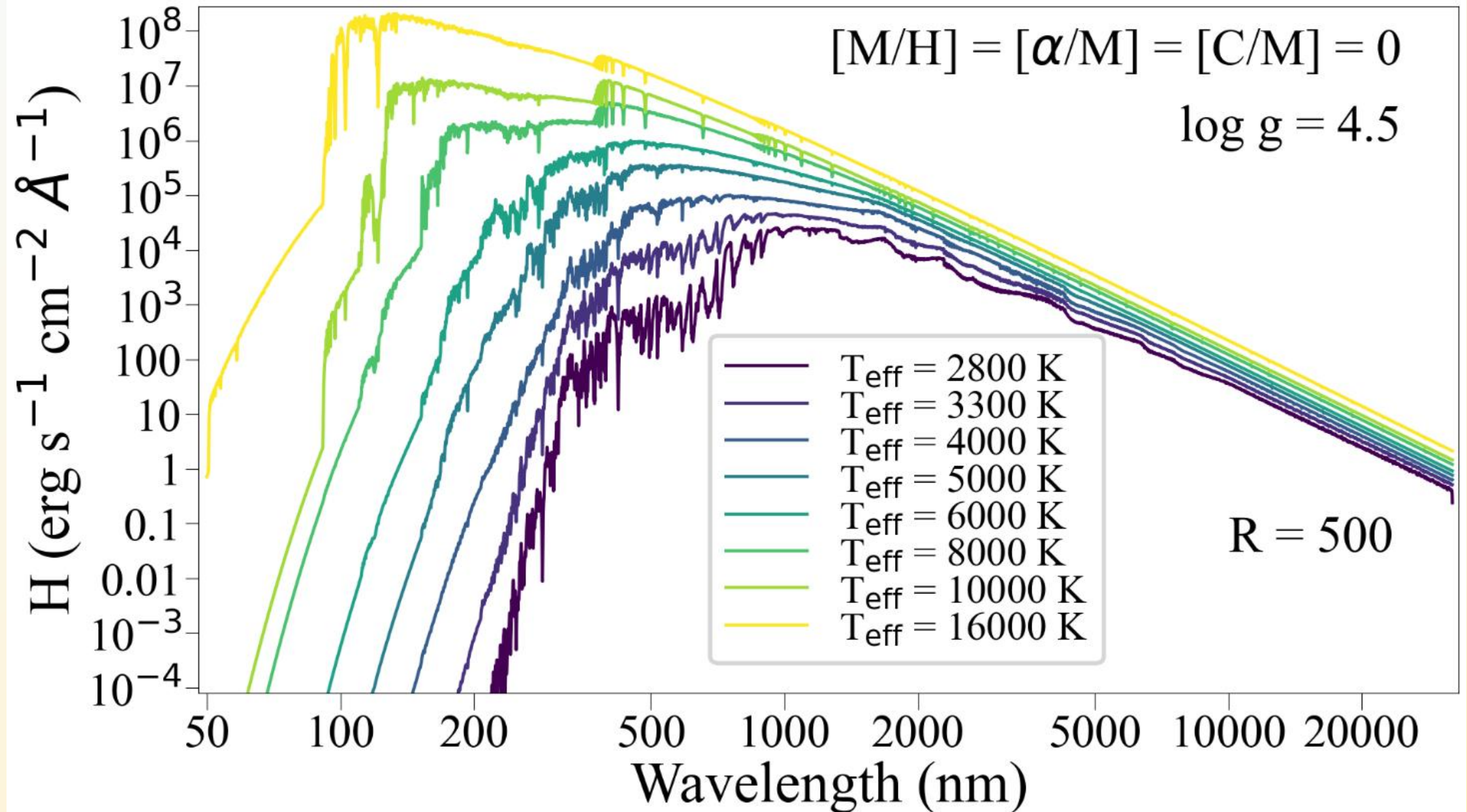


Spectral classification

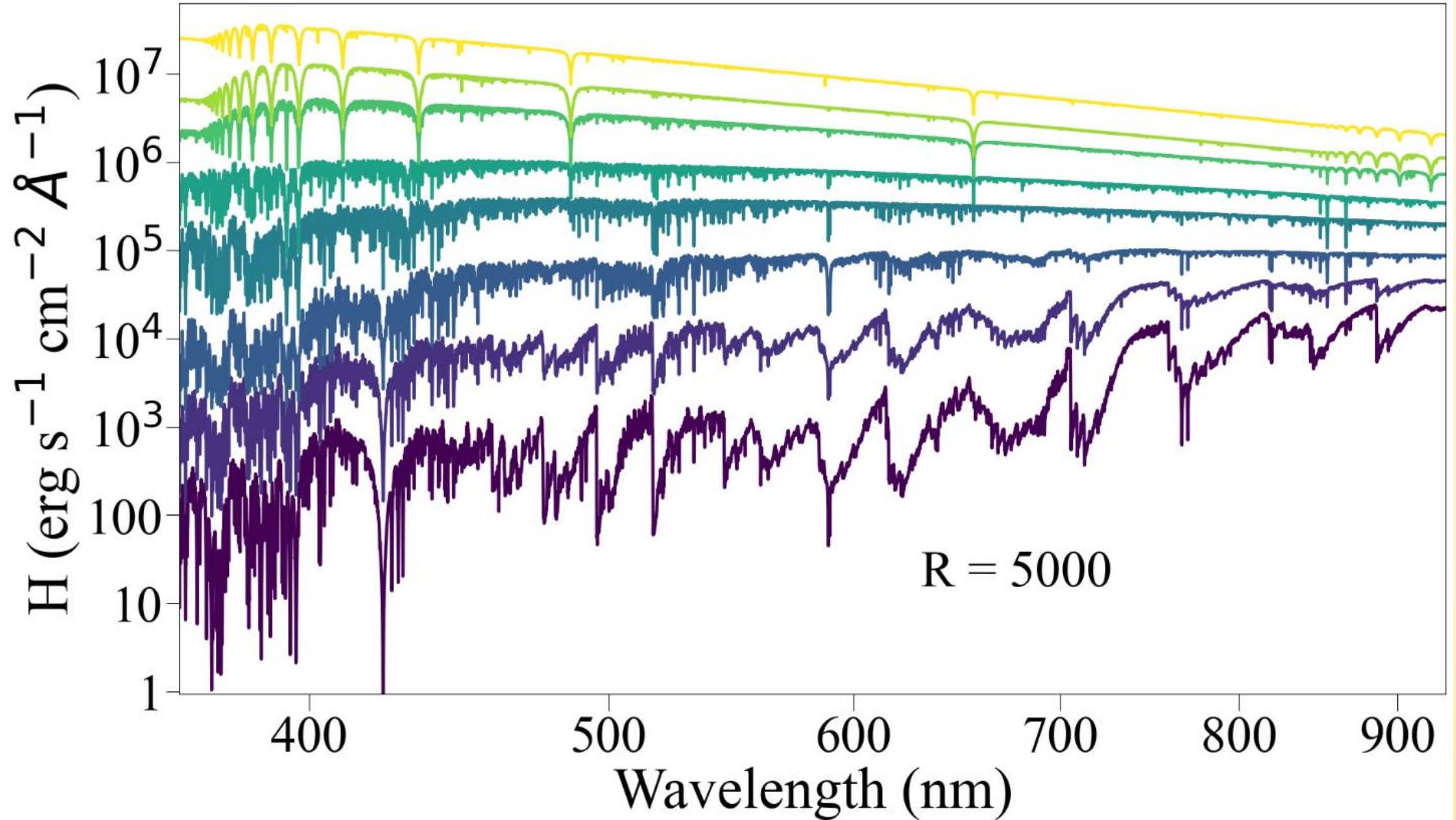
STELLAR CLASSIFICATION (MAIN-SEQUENCE)



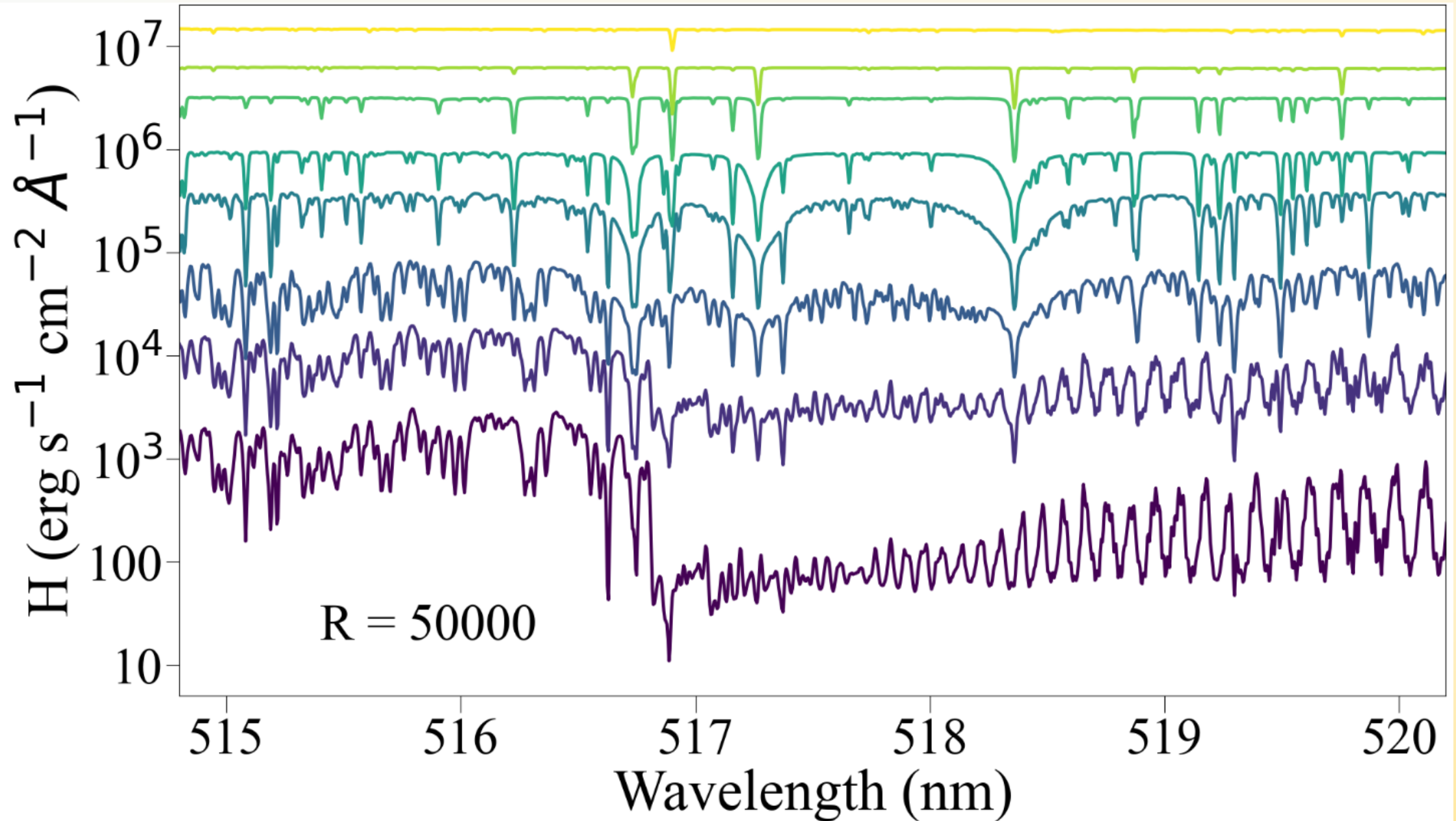
Examples of synthetic spectra (BOSZ)



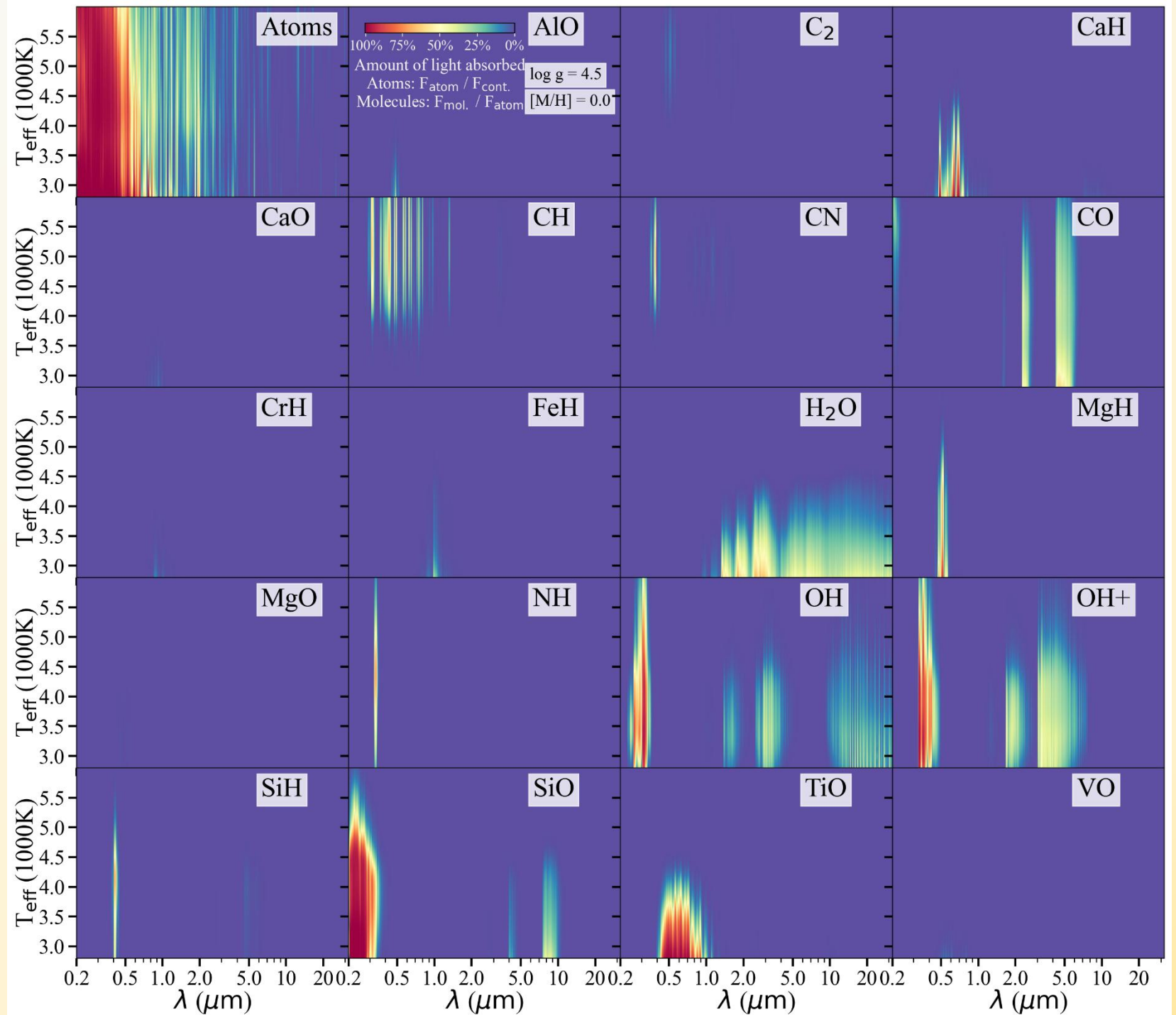
Examples of synthetic spectra



Examples of synthetic spectra



Molecules in the atmosphere



$T_{\text{eff}} = 5750 \text{ K}$, $\log g = 4.5$, $[M/H] = 0.0$

Wavelength

380 nm

440 nm

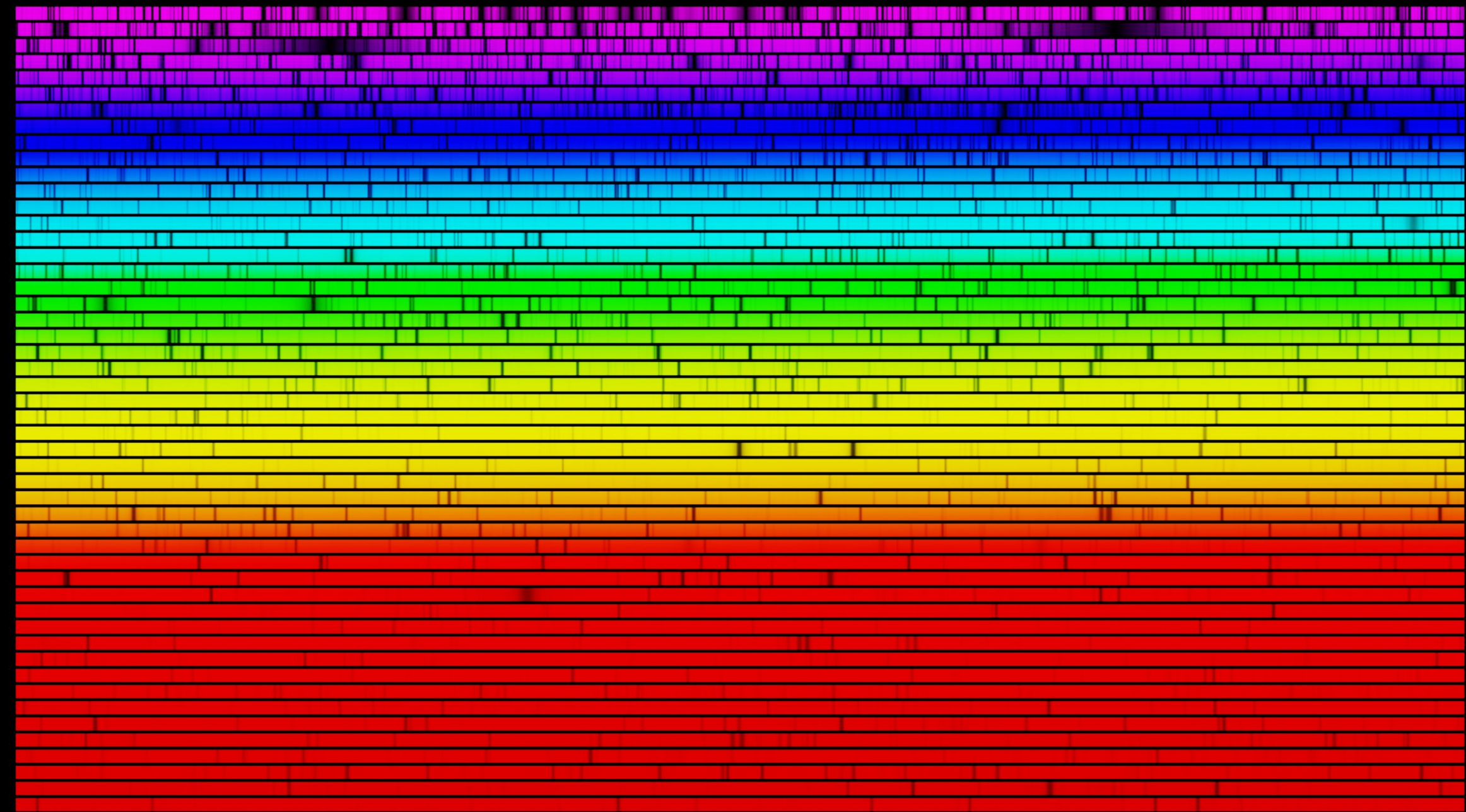
490 nm

510 nm

580 nm

645 nm

760 nm



$T_{\text{eff}} = 3500 \text{ K}$, $\log g = 4.5$, $[M/H] = 0.0$

Wavelength

380 nm

440 nm

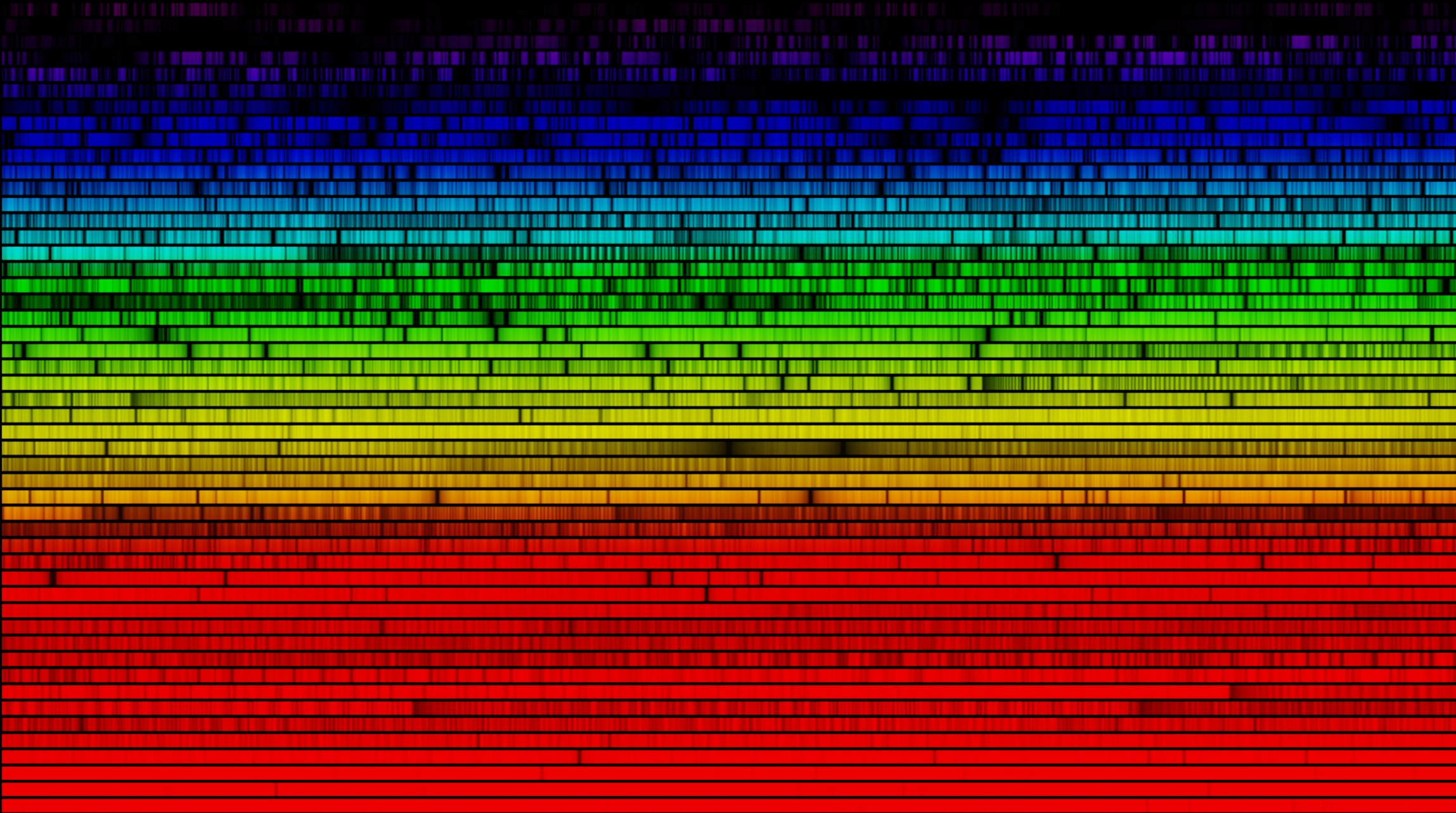
490 nm

510 nm

580 nm

645 nm

760 nm



$T_{\text{eff}} = 2800 \text{ K}$, $\log g = 4.5$, $[M/H] = 0.0$

Wavelength

380 nm

440 nm

490 nm

510 nm

580 nm

645 nm

760 nm